Taylor Tam

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Education

Stanford University — B.S. Computer Science (Systems Track), Minor in Electrical Engineering | *Expected June 2028* | 3.7 GPA
Courses: Operating Systems & Principles, Compilers, Computer Organization & Systems, Infrastructure at Scale, Digital System Design, Probability and Statistics, Applied Matrix Theory, Machine Learning (Grad-level), Deep Learning for Computer Vision (Grad-level), Reinforcement Learning (Grad-level)

Skills: C++, Python, C/Obj-C, Rust, Java, Verilog, SQL | Linux, Docker, Git, GCP, CI/CD, cloud logging | CUDA, PyTorch, OpenCV, RLlib

Experience

Apple — *SWE Intern (iCloud Privacy Team), June 2026 - Present* | Java, Objective C

- Built and launched an **OHTTP token-refund service** using **blind-RSA**, enabling devices to recover tokens lost during signing-key rotations.
- Extended an **Obj-C iOS daemon to identify and flush expired tokens** and **generate blind token replacement requests**
- Developed **cross-repo protobuf APIs** and a **Java/Guice backend** for token verification and secure reissuance.
- Added unit test coverage, staged rollout onto Apple's Private Cloud Compute infrastructure.

Aida — *SWE Intern, June 2025 - September 2025* | Python, SQL, GCP, TypeScript

- Built **distributed Python services** for CRM data pipelines, meeting prep automation, and email entity search, enabling data flow across Google Calendar, internal microservices, and production data systems
- **Automated operational workflows** for deal tracking across CRMs, implemented pipeline filtering, integrated Gemini
- Improved backend reliability by **resolving production-critical issues**, implementing field-stage dependency handling
- **On-call lead** responsible for **deployments, hotfixes, canary releases**; monitored GCP logs/queues, triaged user-facing issues

Stanford Center for Artificial Intelligence and Imaging — *Software Intern, May 2022 - September 2023* | Python, PyTorch, ML

- Trained **ResNet-50 CNN** via transfer learning to detect medical devices in chest X-rays, achieving **97% accuracy**
- Presented at Stanford AI Symposium, authored publication

The Melcher Lab, Stanford University — *Software Engineer, May 2020 - September 2025* | Python, GCP, Swift, Firebase, ML, C#

- Built **GPU inference system** exposed through a REST API and iOS client for real-time clinical image processing
- Developed **Python image-processing and PyTorch UNet model** for liver steatosis quantification
- Used Google Cloud Functions to automate and scale processing workflows across 2,000+ liver donor cases
- Co-authored publication describing a GPU-accelerated embedded system **reducing tissue-analysis latency by 100x**

Projects & Research

From-Scratch COOL Compiler — *March 2026 - June 2026* | C++, Compilers, Flex, Bison

- Built an end-to-end COOL compiler implementing **lexical analysis**, **parsing**, **AST construction**, **semantic analysis**, and **code gen**
- Impl. **inheritance-graph validation, scoped symbol envs**, **method and attribute lookup**, **static dispatch,** and AST type annotation
- Developed **test suites** covering malformed programs, escape sequences, operator precedence, recoverable syntax errors
- Diagnosed **correctness issues** across stages using targeted **test cases** and **intermediate representations**

Rust Based Shape-Safe Deep Learning Library for Apple Silicon — *January 2025 - June 2025* | Rust, Metal Shading Lang

- Built backend-agnostic computer vision library in Rust, w/ compile-time tensor shape validation and hardware acceleration
- Designed **a type-level tensor system** using Rust const generics for compile-time shape enforcement
- Designed zero-cost Rust abstractions supporting interchangeable CPU and GPU execution backends
- Integrated **Metal Performance Shaders (MPS)** on Apple Silicon to accelerate workloads w/ fused GPU kernels
- Benchmarked MPS matrix multiplication and batched MNIST inference on an M1 Pro. Within 15% of PyTorch MPS on GEMM

FPGA Polyphonic Music Synthesizer — *Jan 2026 - March 2026* | Verilog, FPGA, VGA, Audio DSP

- **Verilog FPGA audio pipeline** w/ **ROM playback**, 3-voice chords, ADSR, harmonics, echo, stereo panning, VGA note display
- **Wrote, verified audio/display modules** w/ **waveform testbenches** for chord-mixing, sample-timed envelopes, echo saturation

Multi-Agent Coordination for Flow-SUMO Vehicle Platoons — *Jan 2026 - March 2026* | Python, RLlib, Flow–SUMO, PyTorch

- Implemented **centralized training–decentralized execution (CTDE)**, **independent PPO** agents w/ **RLlib, Flow–SUMO** simulating multi-lane platooning and V2X communication
- Engineered **continuous-control policies** for acceleration and lane selection; evaluated emergent stability